Secilia Tarigan

Bogor, Indonesia | secitarigan@gmail.com | +6285930454198| linkedin.com/in/seciliatrgn

Fresh graduate in Telecommunication Engineering from ITB with hands-on experience across telecommunications, network infrastructure, cybersecurity, cloud infrastructure, machine learning, and digital technologies. Strong technical foundation in C++, Python, networking, Linux, network security, cloud platforms, and system simulation, complemented by experience leading technical teams and delivering hands-on technical training.

Education

Institut Teknologi Bandung — Bandung, Indonesia
Bachelor of Telecommunication Engineering — Aug 2022 – current
• Awards: Ganesha Karya Award (2025), Hana Nanum Foundation Scholarship (2025, 2026).

Projects

5G Non-Terrestrial Network (NTN) Random Access Optimization (Bachelor Thesis) (2026):
• Developed and evaluated a Group-Based Preamble Allocation algorithm to improve random access performance for massive IoT devices in 5G Non-Terrestrial Networks (NTN).
• Implemented and modified MAC-layer procedures in 5G-LENA (ns-3) using C++ to model PRACH, contention-based random access, and LEO satellite communication.

Software-Defined Doppler Radar System (2025):
• Developed a real-time bistatic radar using USRP B210 and LabVIEW that measures target velocities by implementing a Direct Path Reference synchronization to cancel phase noise.
• Implemented a DSP pipeline featuring Conjugate Mixing and FFT analysis, validating performance across 3 GHz and 6 GHz carrier frequencies to enhance Doppler sensitivity.

"KiraUangKU" AI Currency Detector (2025):
• Developed a full-stack web application for automated banknote recognition using a YOLOv8 model trained on a currency-specific image dataset.
• Architected a Next.js dashboard integrated with a Flask API backend for real-time image processing, improving detection accessibility for users.

IoT Smart Lighting System (2024):
• Created a motion-activated lighting system featuring remote control capabilities via Telegram Bot API, enhancing energy efficiency through automated occupancy detection.
• Integrated PIR sensors and established sensor-to-cloud connectivity to provide users with seamless real-time control over lighting environments.

Smart Home Intrusion Detector (Machine Learning) (2024):
• Engineered an automated intrusion detection system for IoT environments using Random Forest and Gradient Boosting algorithms to identify malicious network patterns.
• Achieved a 99.51% classification accuracy and 0.9997 ROC-AUC score by training models on a massive dataset of 148,517 entries.
• Implemented a robust data pipeline using Python (Scikit-learn & Pandas), featuring Label Encoding for categorical features and 70/30 data splitting to ensure model reliability and minimize overfitting.
• Optimized detection performance, successfully reducing False Positives and False Negatives to under 1% for real-time security monitoring.

Security Integration & Monitoring (Wazuh & pfSense) (2024):
• Designed and deployed a multi-layered secure network architecture by integrating pfSense firewall with the Wazuh security monitoring platform.
• Configured real-time threat detection and centralized log analysis, enabling Suricata IDS to provide deep visibility into network events.
• Implemented enterprise-grade security features including File Integrity Monitoring (FIM) and automated vulnerability scanning across client machines.

Skills & Achievements

Programming: C++, Python, JavaScript, MATLAB
Telecommunications & Networking: 5G/5G NTN, Network Fundamentals, SD-WAN, MPLS, Fiber Optic, PRACH/RACH
Cybersecurity: pfSense, Wazuh, Security Onion, Suricata, Network Monitoring, Digital Forensics
Cloud & Infrastructure: Google Cloud Platform (GCP), AWS Academy, Docker, VMware/KVM
Machine Learning: YOLOv8, Scikit-learn, Pandas
Tools: ns-3/5G-LENA, LabVIEW, USRP B210, Git/GitHub, Linux/WSL
Certifications:
• Set Up an App Dev Environment on Google Cloud – Google Cloud Skill Badge, Issued Apr 2025.
• Implement Load Balancing on Compute Engine – Google Cloud Skill Badge, Issued Mar 2025.
• APIE e-Workshop – SOI Asia (School on Internet Asia), Issued Nov 2025.
• Understanding the Internet – SOI Asia (School on Internet Asia), Issued Nov 2025.

• Operating the Internet – SOI Asia (School on Internet Asia), Issued Nov 2025.

Achievements:

• **Ganesha Karya Award** — 2025

• **Hana Nanum Foundation Scholarship** — 2025, 2026

• **The Best Institutional Integration Solution – ICON 3.0 (Innovation Competition in Logistics Business Case Solution).**

Languages: Indonesian (Native), English (Professional Working Proficiency).

Work Experience

Institut Teknologi Bandung Bandung, Indonesia

Teaching Assistant - Telecommunication Network Security Aug 2025 - Dec 2025

• Mentored 45 students in deploying integrated network security infrastructures using pfSense, Wazuh, and Security Onion, strengthening practical skills in network monitoring and intrusion detection.

• Led hands-on training covering Next-Generation Firewall (NGFW) technologies, including Suricata, pfBlockerNG, and ntopng, and supervised digital forensic analysis using Autopsy.

• Guided students through practical implementation, troubleshooting, and analysis of network security monitoring environments.

Institut Teknologi Bandung Bandung, Indonesia

Teaching Assistant - Signal Processing in Discrete Time Aug 2025 – Dec 2025

• Led MATLAB-based laboratory sessions covering the practical design and implementation of digital filter architectures.

• Facilitated tutorials for 22 students on Z-Transform, DFT, and FIR/IIR digital filters, supporting students in applying signal-processing concepts to technical problems.

• Managed quiz grading and laboratory documentation with 100% accuracy, ensuring compliance with academic and curriculum standards.

Pelindo Digital Solution Jakarta, Indonesia

Digital Infrastructure and Security Intern Jul 2025 – Aug 2025

• Developed technical planning and Bill of Quantity (BoQ) for Palo Alto SD-WAN deployments using PA-1410, PA-440, and Panorama M-300 for multiple regional sites.

• Designed branch network topologies incorporating High Availability (HA) and hybrid MPLS/Internet connectivity to support resilient corporate communications.

• Developed Business Continuity Management (BCM) policies based on ISO 22301:2019, including Business Impact Analysis (BIA), Recovery Time Objective (RTO), and Recovery Point Objective (RPO).

• Conducted a site survey at IPC TPK Tanjung Priok 2 and contributed to the design of a redundant Fiber Optic Ring Backbone, including strategic deployment planning for 39 Wireless Access Points.

Organizational Experience

IMT "Signum" ITB (Telecommunication Engineering Student Association) Bandung, Indonesia

Head of Media, Communication, and Information Department Apr 2025 - Mar 2026

• Led 3 specialized divisions (Graphic Design, Publication, and Documentation) to strengthen organizational branding and digital strategy.

• Engineered a social media strategy focused on educational telecommunications content, achieving a 30% increase in audience engagement.

Head of Media Department for SPEKTRUM IMT "Signum" ITB Aug 2024 - Oct 2024

• Led the Media Department for SPEKTRUM, an introductory study of telecommunications for the Signum family. Successfully coordinated the Publication & Documentation and Graphic Design divisions to ensure effective communication and event branding.

ITB Jazz Bandung, Indonesia

Content Strategist Oct 2023 – Jun 2024

Creative Marketing for Ganesha Session 2023 event Mar 2023 – Jun 2023

• Architected creative marketing campaigns for flagship events including "Ganesha Session 2023" executed multi-channel publication plans to increase event visibility, contributing to the successful promotion.